

Introduction to Chromosome Simulations



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Genome Organization

Length of the human genome

6.4×10^9 base pairs



1 bp $\Rightarrow$ 0.34 nm  $\sim$ 2m long



Cell nucleus $\sim 10 \mu\text{m}$



Genome Organization

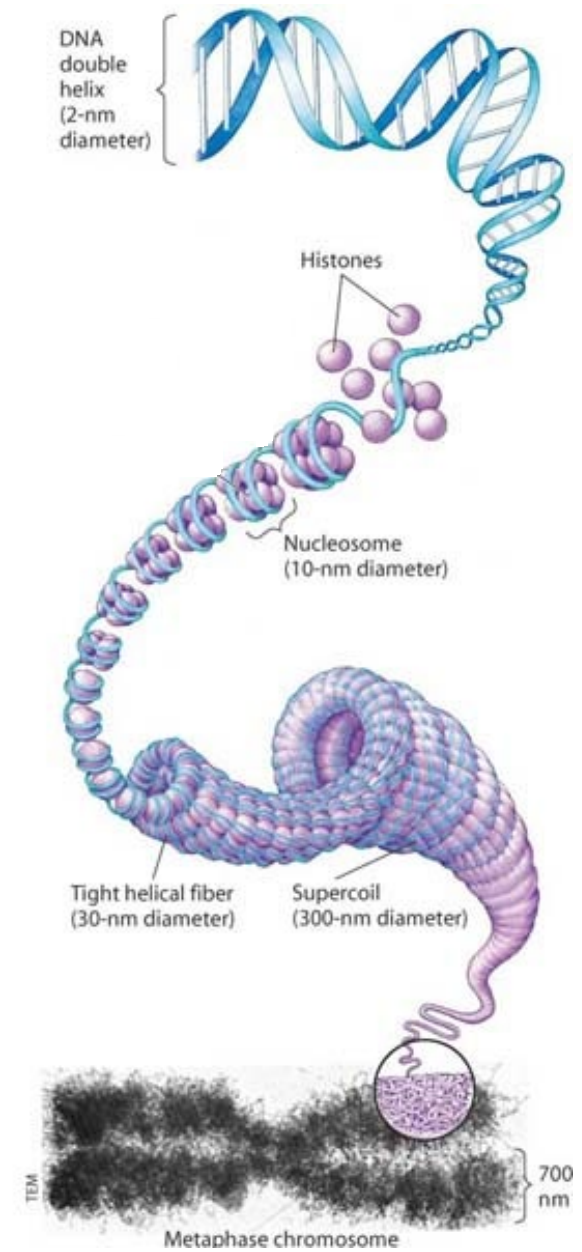
Length of the human genome

6.4×10^9 base pairs



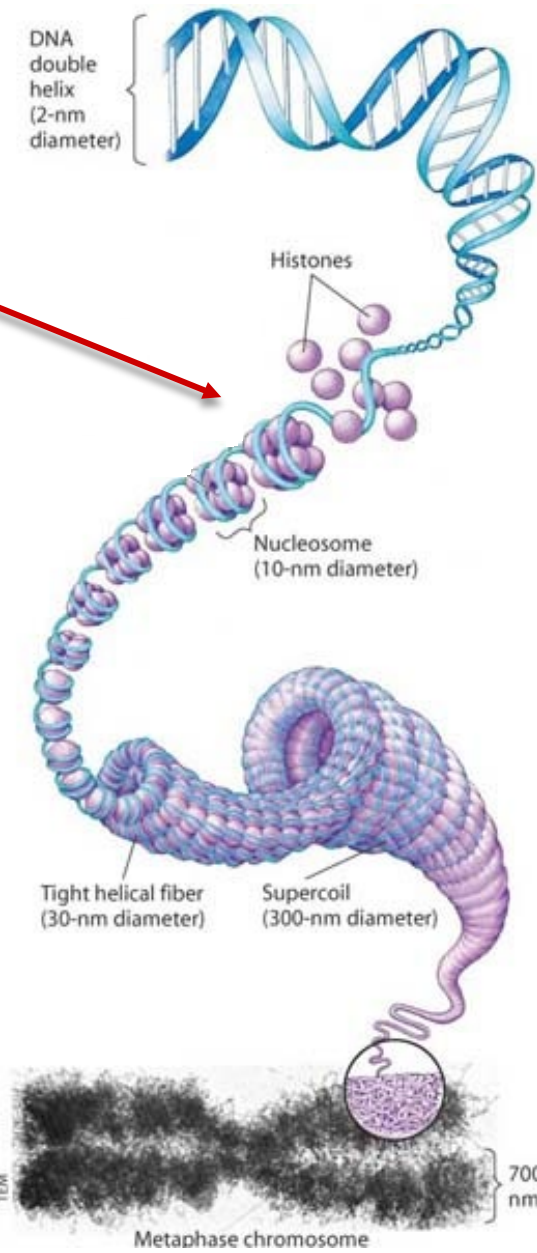
1 bp $\Rightarrow$ 0.34 nm $\rightarrow$ ~ 2m long

Cell nucleus ~ 10 μ m



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Luger, K. et al. Crystal structure of the nucleosome core particle at 2.8 Å resolution. *Nature* **389**, 252 (1997)

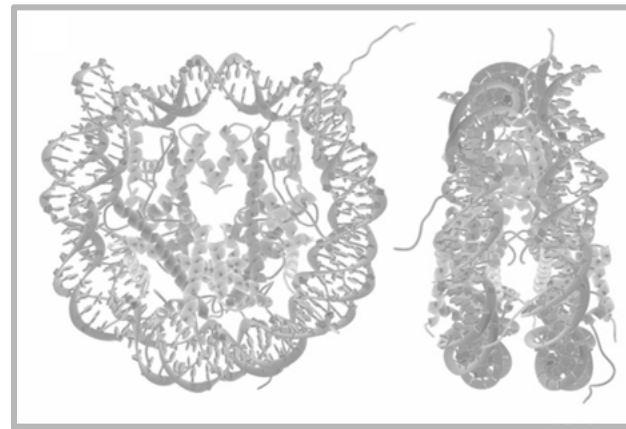


1 bp $\Rightarrow$ 0.34 nm $\longrightarrow$ ~ 2m long

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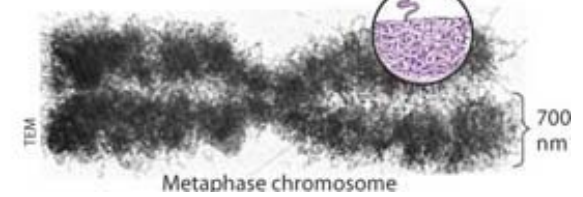


Histones

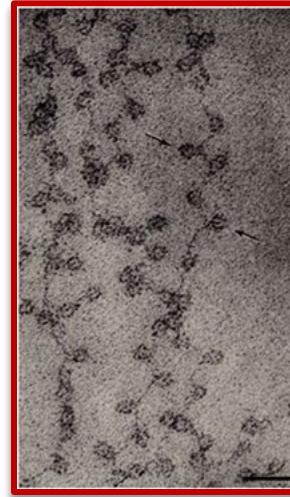
Nucleosome (10-nm diameter)

Tight helical fiber (30-nm diameter)

Supercoil (300-nm diameter)

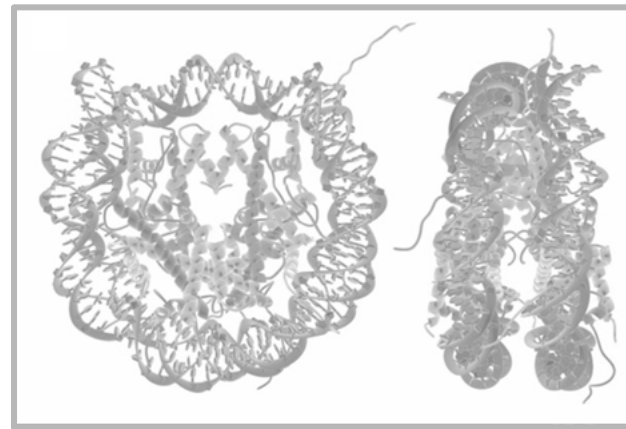


Olins, D. E. & Olins, A. L. Chromatin history: our view from the bridge. *Nature Reviews Molecular Cell Biology* 4, 811 (2003)



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Luger, K. et al. Crystal structure of the nucleosome core particle at 2.8 Å resolution. *Nature* 389, 252 (1997)

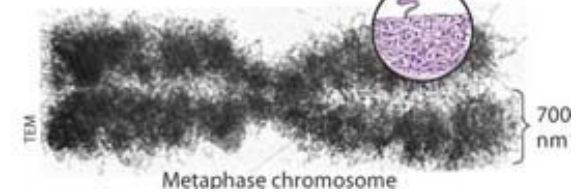


Histones

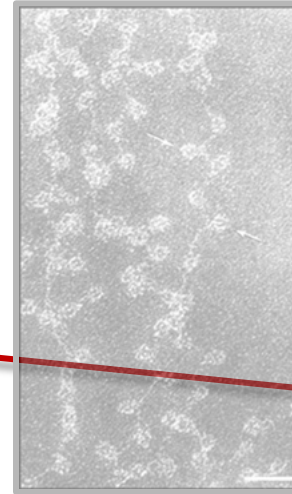
Nucleosome (10-nm diameter)

Tight helical fiber (30-nm diameter)

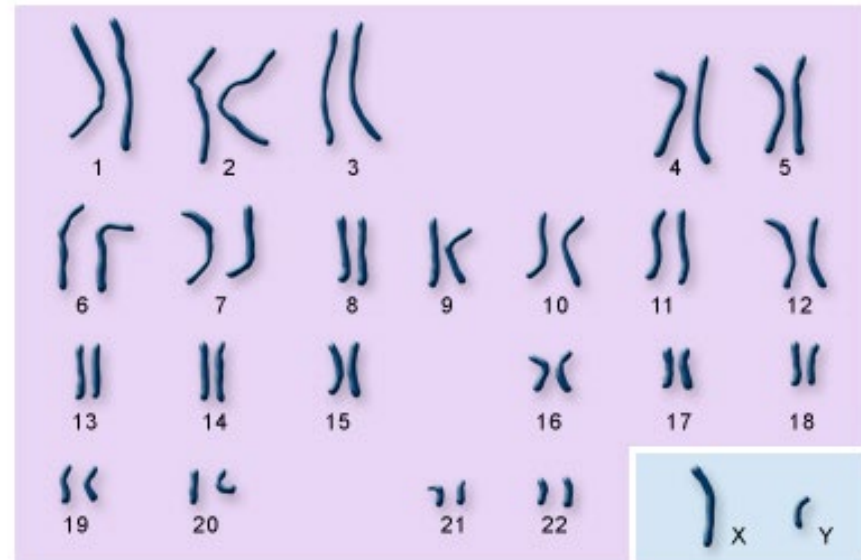
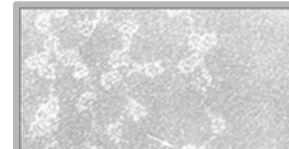
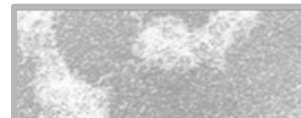
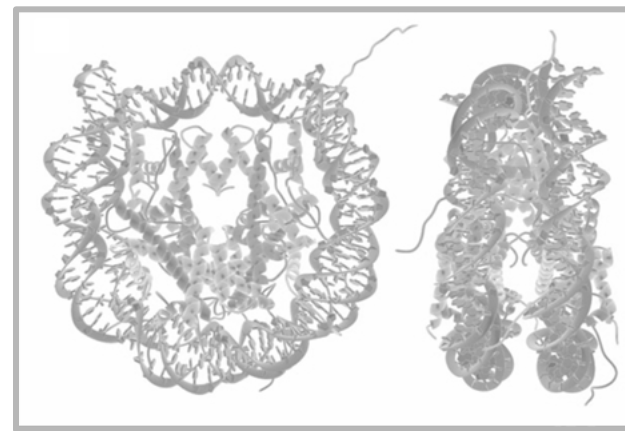
Supercoil (300-nm diameter)



Olins, D. E. & Olins, A. L. Chromatin history: our view from the bridge. *Nature Reviews Molecular Cell Biology* 4, 811 (2003)



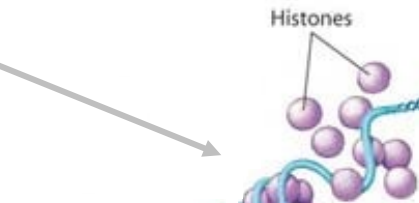
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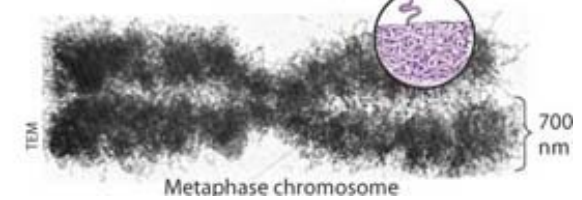
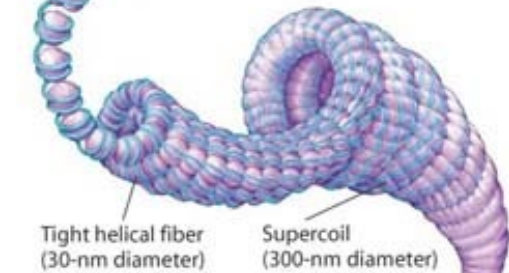
autosomes

sex chromosomes

U.S. National Library of Medicine



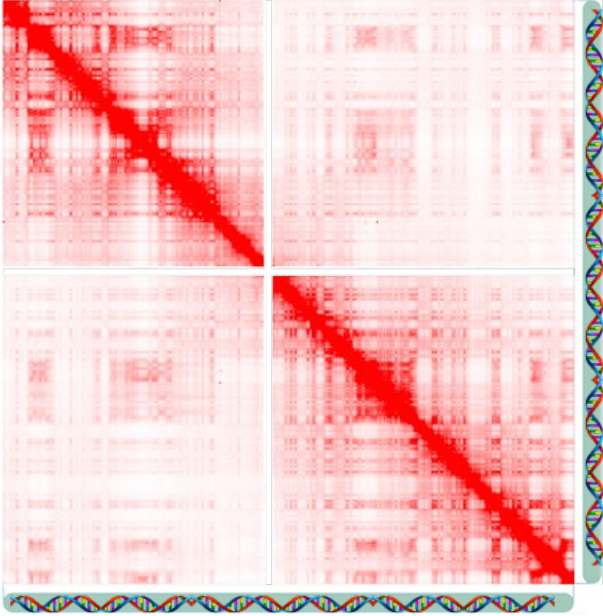
Nucleosome (10-nm diameter)



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Hi-C

Chromosome “Contact Map”



Interphase - 100 Kb resolution

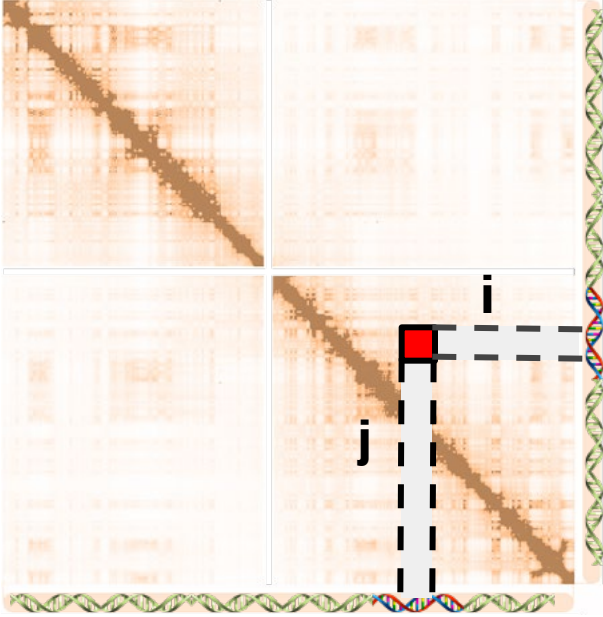
Human - Chromosome 3 - 180Mb

GM12878 cell line

Lieberman-Aiden E., Dekker J. et al. *Comprehensive mapping of long-range interactions reveals folding principles of the human genome*. Science. 2009.

Hi-C

Chromosome “Contact Map”



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Chromosome “Contact Map”

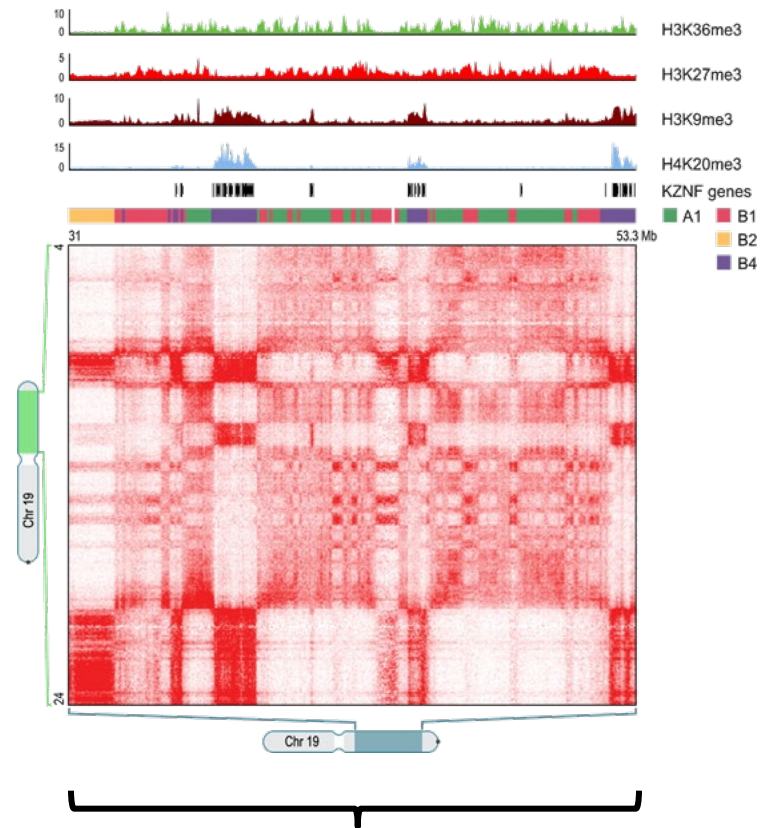


Interphase - 100 Kb resolution

Human - Chromosome 3 - 180Mb

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Lieberman-Aiden E., Dekker J. et al. *Comprehensive mapping of long-range interactions reveals folding principles of the human genome*. Science. 2009.



Chromatin Types: A and B

Rao S, Lieberman-Aiden E., et al. *A 3D Map of the Human Genome at Kilobase Resolution Reveals Principles of Chromatin Looping*. Cell. 2014

Hi-C

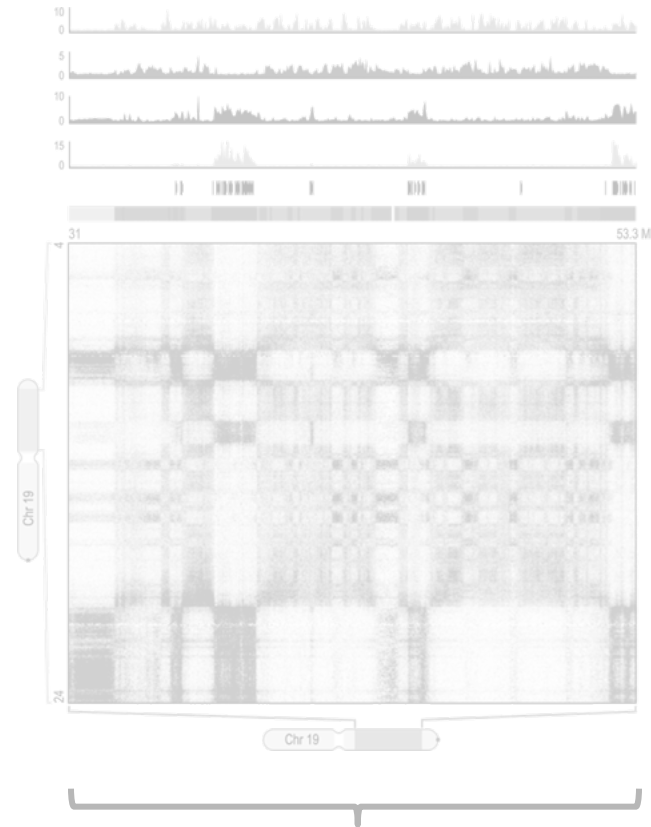
Chromosome “Contact Map”



Interphase - 100 Kb resolution

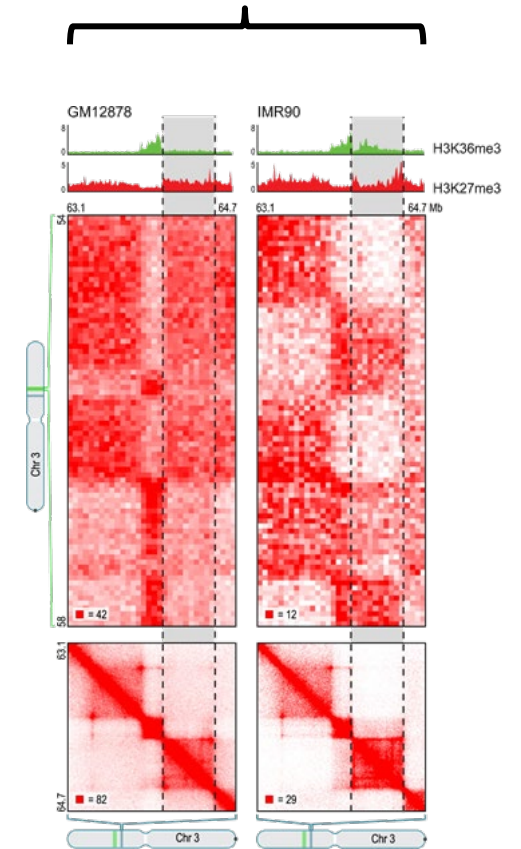
Human - Chromosome 3 - 180Mb

GM12878 cell line



Chromatin Types: A and B

Different Cell Lines

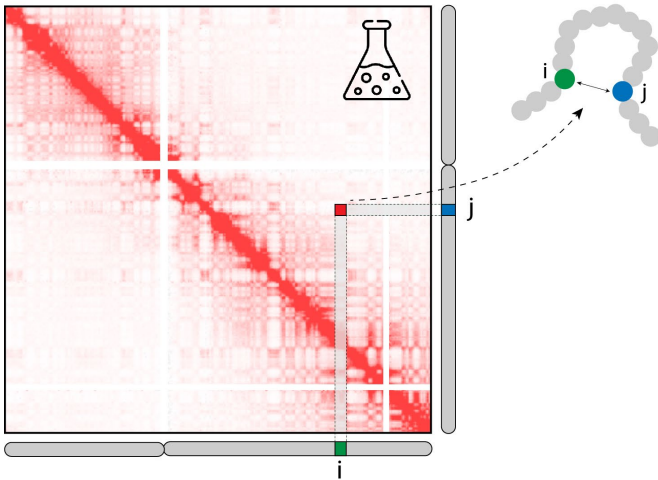


Lieberman-Aiden E., Dekker J. et al. *Comprehensive mapping of long-range interactions reveals folding principles of the human genome*. Science. 2009.

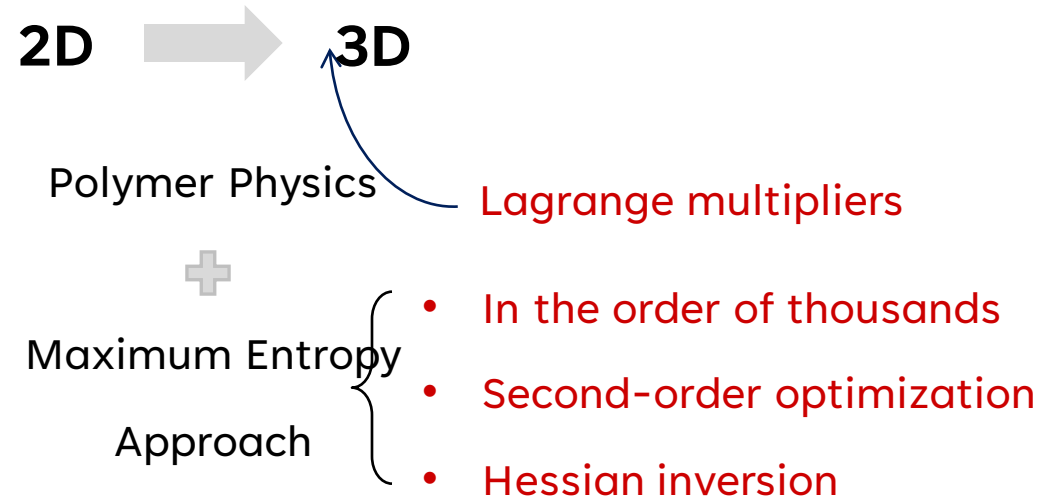
Rao S, Lieberman-Aiden E., et al. *A 3D Map of the Human Genome at Kilobase Resolution Reveals Principles of Chromatin Looping*. Cell. 2014

Maximum Entropy Approach

Experimental Hi-C Map



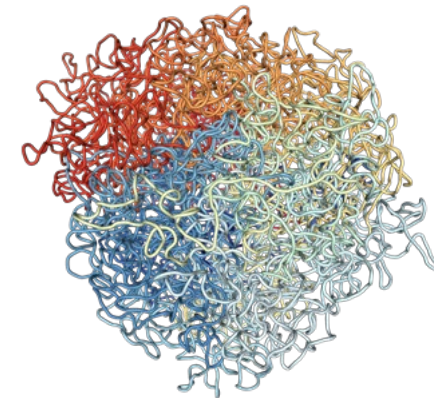
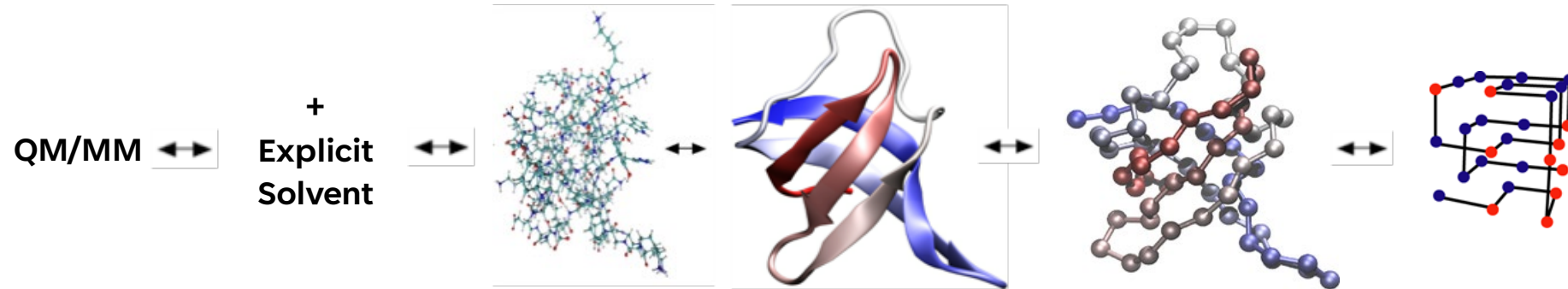
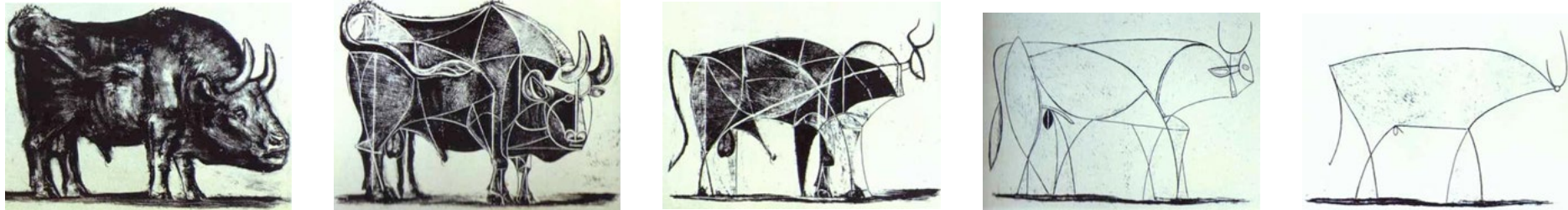
- 2D Heat map
- Exp. uses over 1 million cells
- What about 3D structures?



Zhang, B & Wolynes, P, PNAS, 2015
Di Pierro, M & Zhang, B et al. PNAS 2016
Contessoto, VG, et al. COSB, 2022

Simplified Models

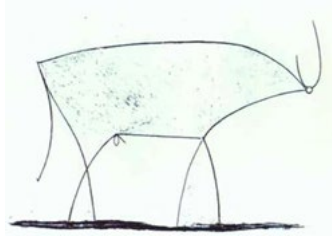
Picasso, El toro (1944-45)



MiChroM – Minimal Chromatin Model

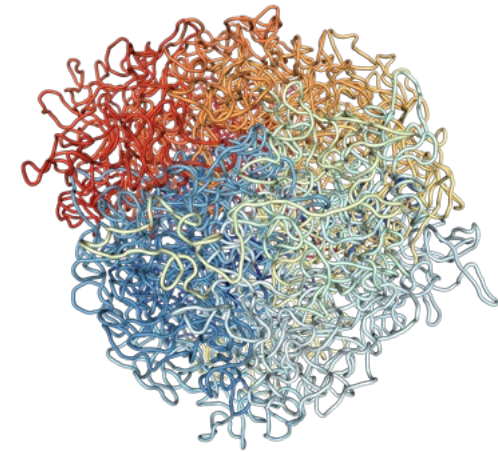
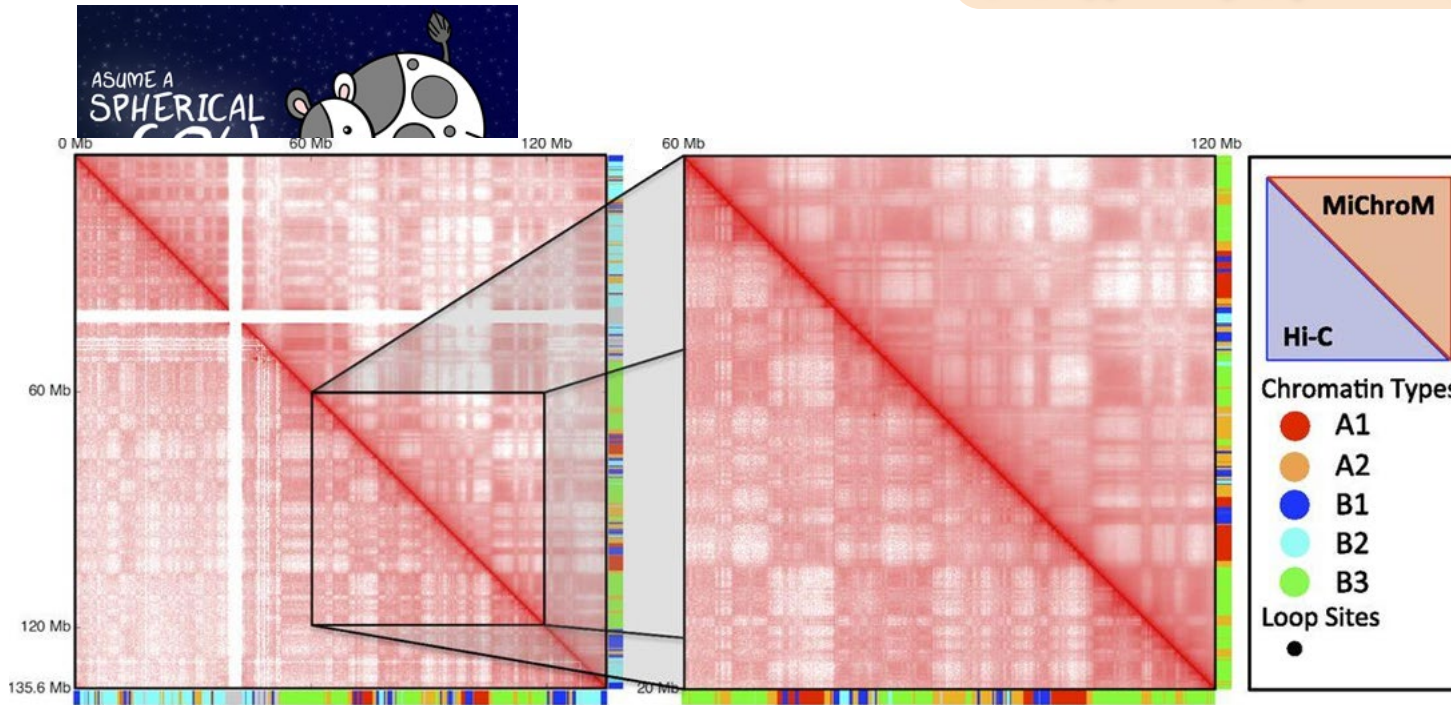
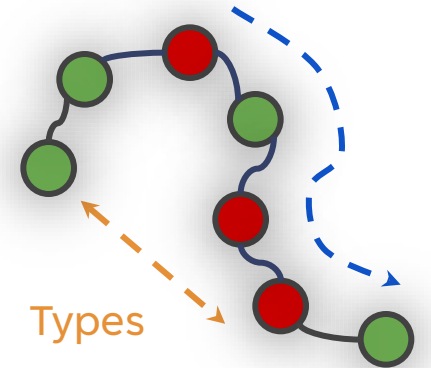


Picasso, El toro (1944-45)



$$U_{\text{MiChroM}}(\vec{r}) = U_{\text{HP}}(\vec{r}) + \sum_{d=3}^{d_{\text{cutoff}}} \gamma(d) \sum_i f(r_{i,i+d}) + \sum_{\substack{k \geq l \\ k, l \in \text{Types}}} \alpha_{kl} \sum_{\substack{i \in \{\text{Loci of Type } k\} \\ j \in \{\text{Loci of Type } l\}}} f(r_{ij})$$

Ideal Chromosome



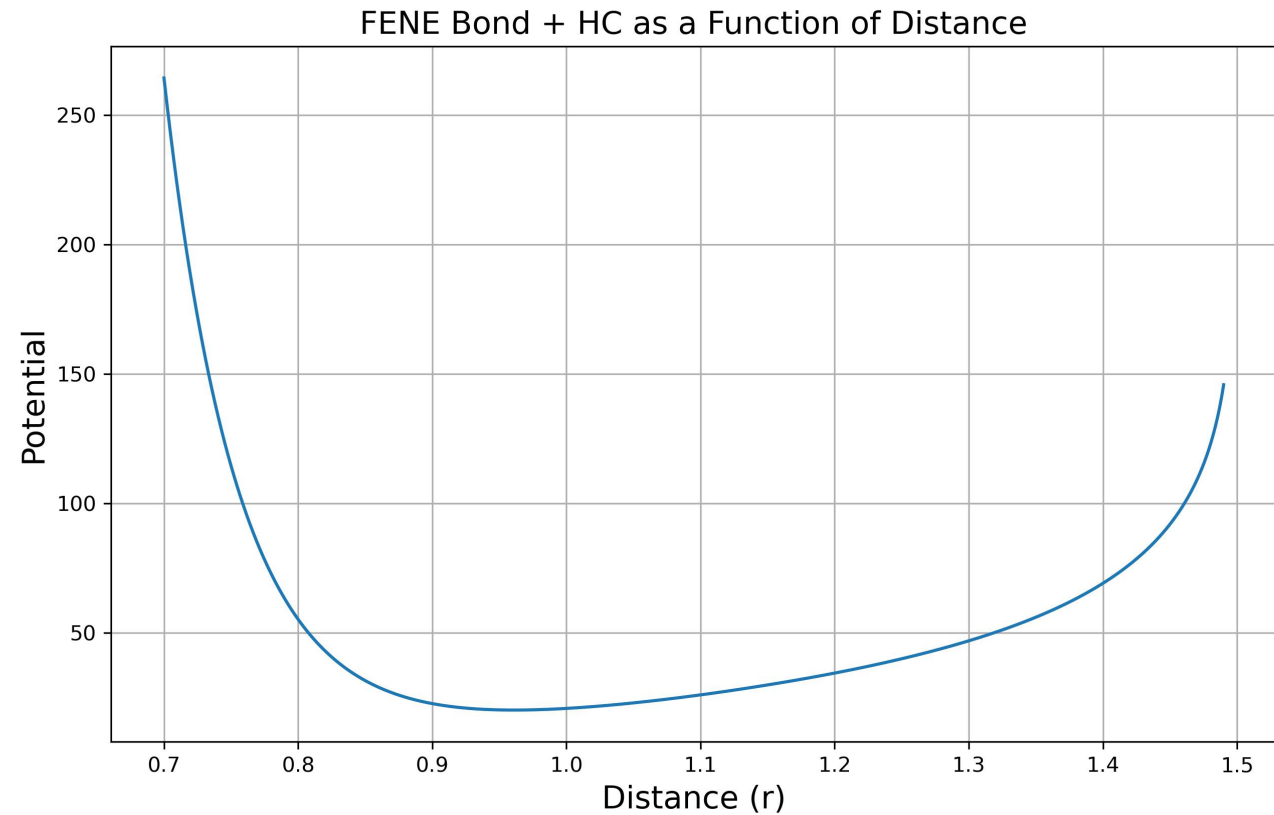
Ensemble of 3D Structures

Potentials

$$U_{HP}(\vec{r}) = \sum_{i \in \{\text{Loci}\}} U_{\text{FENE}}(\vec{r}_{i,i+1}) + \sum_{i \in \{\text{Loci}\}} U_{hc}(\vec{r}_{i,i+1}) + \sum_{i \in \{\text{Angles}\}} U_{\text{Angle}}(\theta_i) + \sum_{\substack{i,j \in \{\text{Loci}\} \\ j > i+2}} U_{sc}(\vec{r}_{i,j})$$

$$U_{\text{FENE}}(\vec{r}_{i,j}) = \begin{cases} -\frac{1}{2} K_b R_0^2 \ln \left[1 - \left(\frac{r_{i,j}}{R_0} \right)^2 \right] & r_{i,j} \leq R_0 \\ 0 & r_{i,j} > R_0 \end{cases}$$

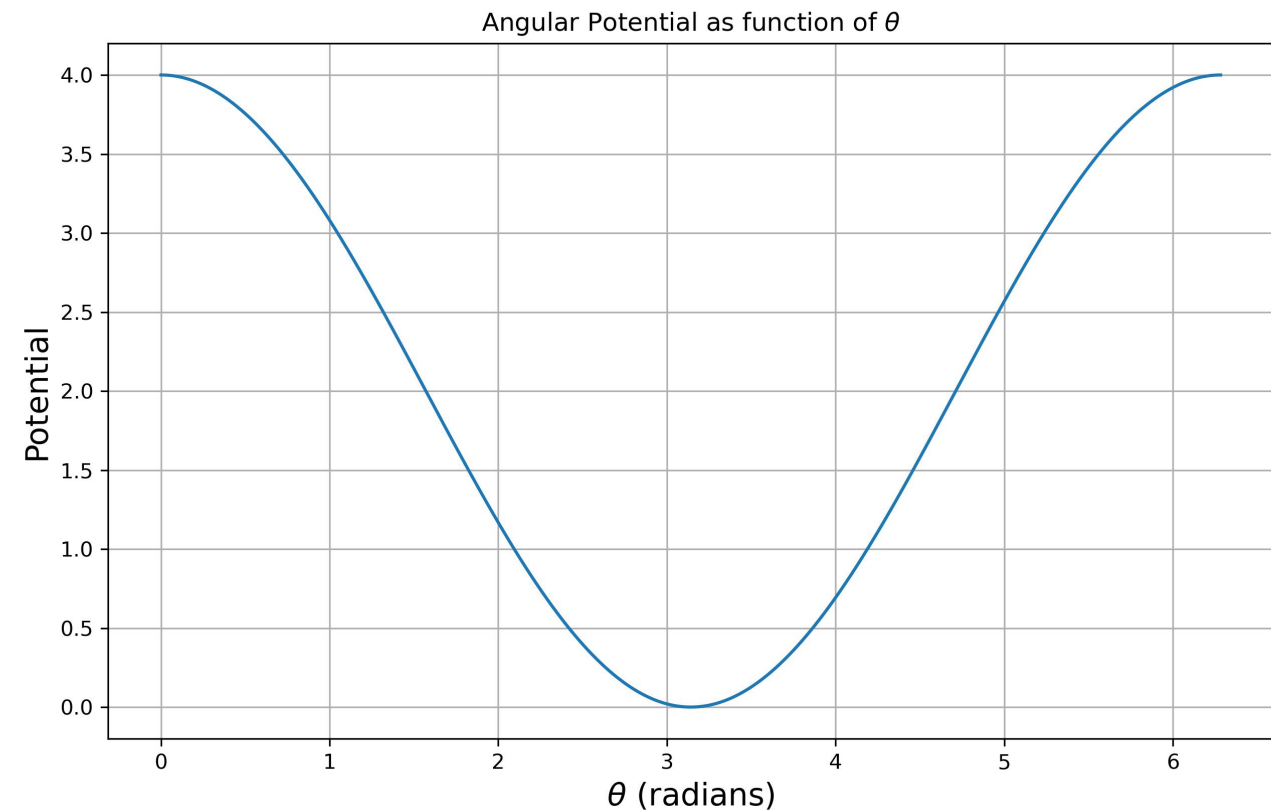
$$U_{hc}(\vec{r}_{i,j}) = \begin{cases} 4\epsilon \left[\left(\frac{\sigma}{r_{i,j}} \right)^{12} - \left(\frac{\sigma}{r_{i,j}} \right)^6 + \frac{1}{4} \right] & r_{i,j} \leq \sigma 2^{\frac{1}{6}} \\ 0 & r_{i,j} > \sigma 2^{\frac{1}{6}} \end{cases}$$



Potentials

$$U_{HP}(\vec{r}) = \sum_{i \in \{\text{Loci}\}} U_{\text{FENE}}(\vec{r}_{i,i+1}) + \sum_{i \in \{\text{Loci}\}} U_{hc}(\vec{r}_{i,i+1}) + \sum_{i \in \{\text{Angles}\}} U_{\text{Angle}}(\theta_i) + \sum_{\substack{i,j \in \{\text{Loci}\} \\ j > i+2}} U_{sc}(\vec{r}_{i,j})$$

$$U_{\text{Angle}}(\theta_i) = K_a[1 - \cos \theta_i - \theta_0]$$

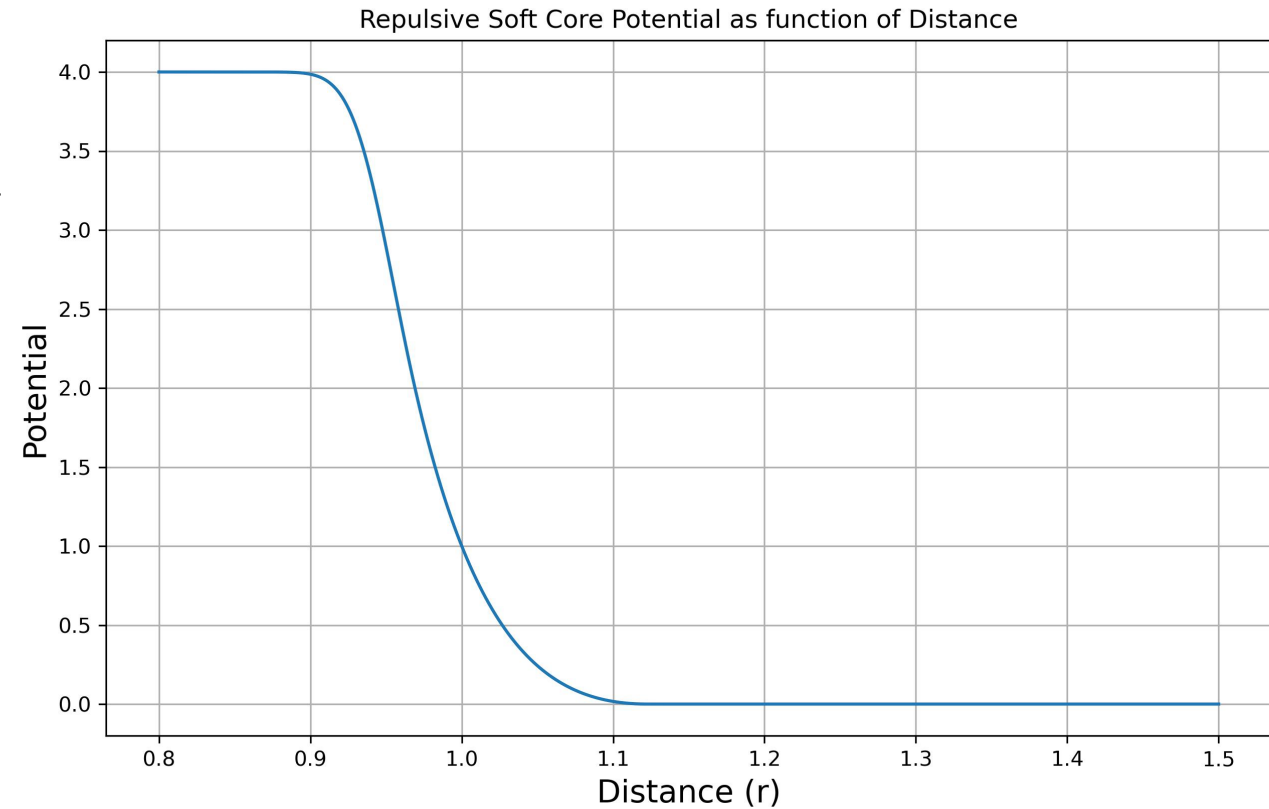


Potentials

$$U_{HP}(\vec{r}) = \sum_{i \in \{\text{Locs}\}} U_{\text{FENE}}(\vec{r}_{i,i+1}) + \sum_{i \in \{\text{Locs}\}} U_{hc}(\vec{r}_{i,i+1}) + \sum_{i \in \{\text{Angles}\}} U_{\text{Angle}}(\theta_i) + \sum_{\substack{i,j \in \{\text{Locs}\} \\ j > i+2}} U_{sc}(\vec{r}_{i,j})$$

$$U_{sc}(\vec{r}_{i,j}) = \begin{cases} \frac{1}{2} E_{\text{cut}} \left[1 + \tanh \left(\frac{2U_{LJ}(\vec{r}_{i,j})}{E_{\text{cut}}} - 1 \right) \right] & r_{i,j} \leq r_0 \\ U_{LJ}(\vec{r}_{i,j}) & r_0 \leq r_{i,j} \leq \sigma 2^{\frac{1}{6}} \\ 0 & r_{i,j} > \sigma 2^{\frac{1}{6}} \end{cases}$$

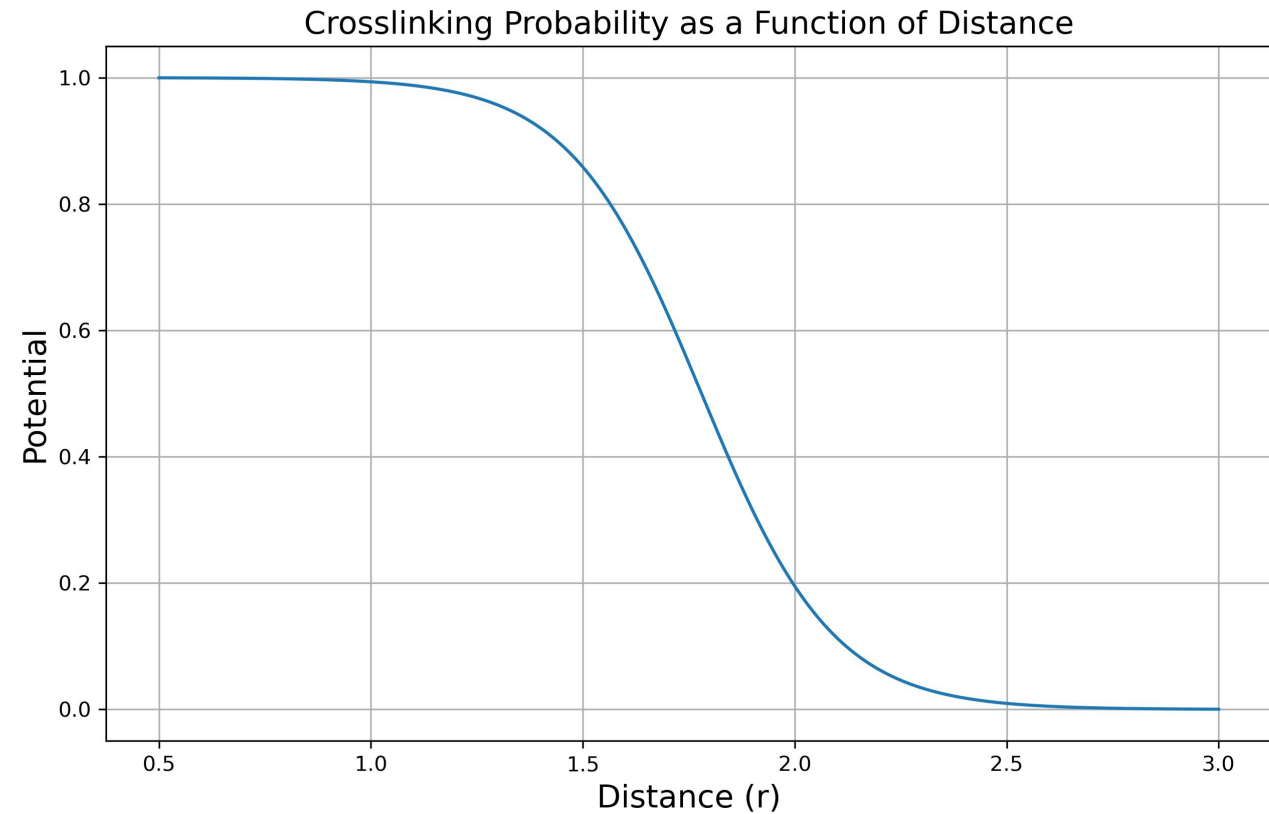
$$U_{LJ}(\vec{r}_{i,j}) = 4\epsilon \left[\left(\frac{\sigma}{r_{i,j}} \right)^{12} - \left(\frac{\sigma}{r_{i,j}} \right)^6 + \frac{1}{4} \right]$$



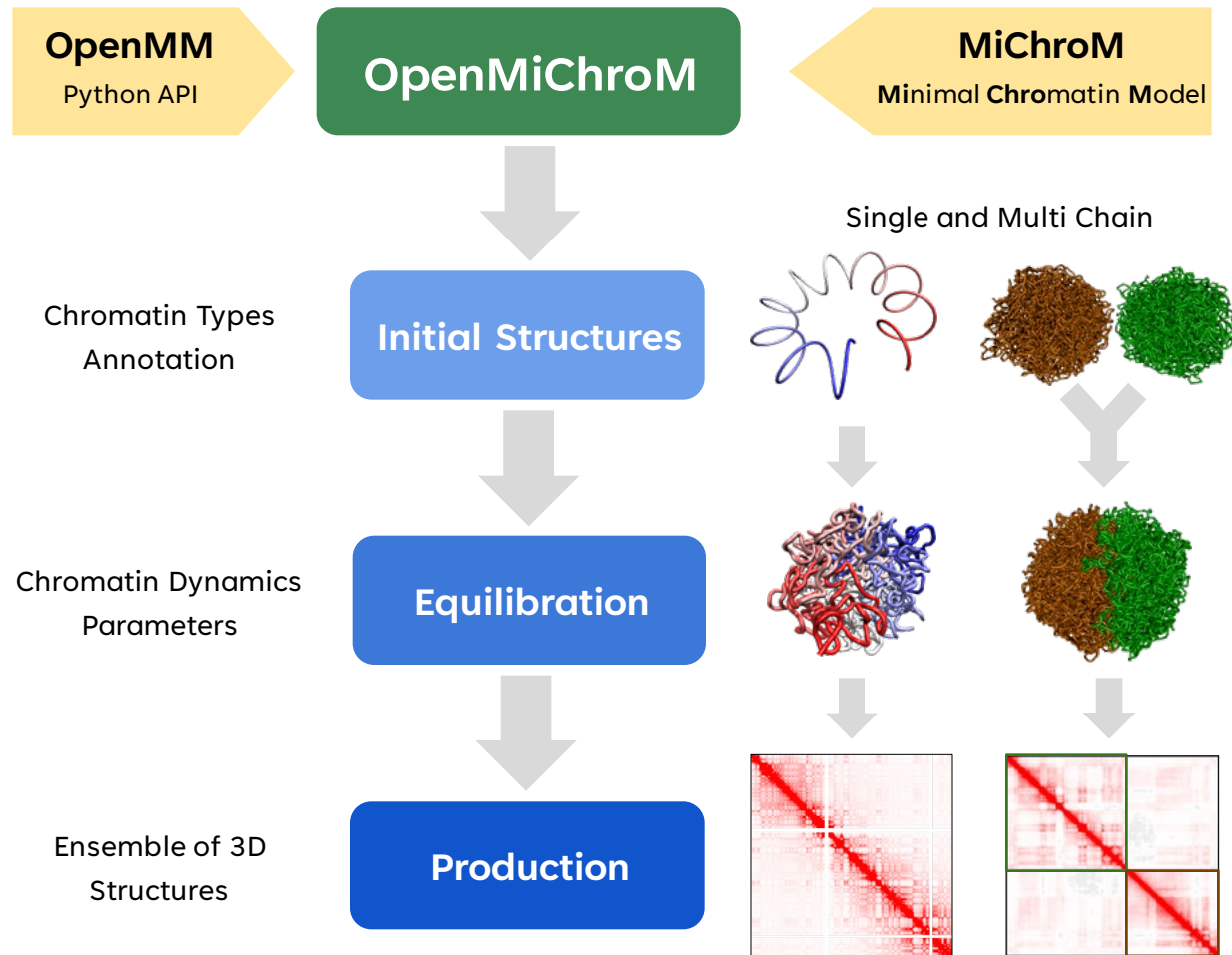
Potentials

$$U_{\text{MiChroM}}(\vec{r}) = U_{HP}(\vec{r}) + \sum_{d=3}^{d_{\text{cutoff}}} \gamma(d) \sum_i f(r_{i,i+d}) + \sum_{\substack{k \geq l \\ k, l \in \text{Types}}} \alpha_{kl} \sum_{\substack{i \in \{\text{Loci of Type } k\} \\ j \in \{\text{Loci of Type } l\}}} f(r_{ij})$$

$$f(r_{i,j}) = \frac{1}{2} (1 + \tanh [\mu(r_c - r_{i,j})])$$

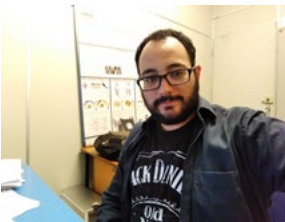
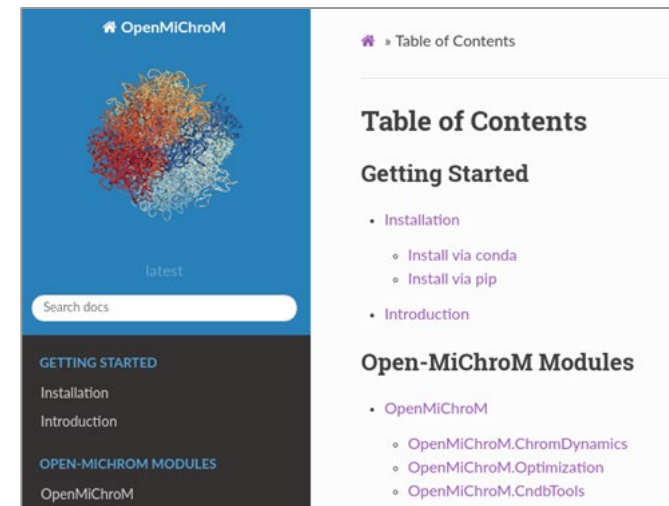


OpenMiChroM



1. Input .bed (Sub)Comp. files from PyMEGABASE
2. Output .cndb trajectory file (Nucleome Data Bank)
3. Analysis performed using cndbTools

```
cndbt = cndbTools()  
cndbt.xyz  
cndbt.traj2HiC
```



Antonio B. Oliveira Jr.

“There is always a bug” – José Onuchic, 2017

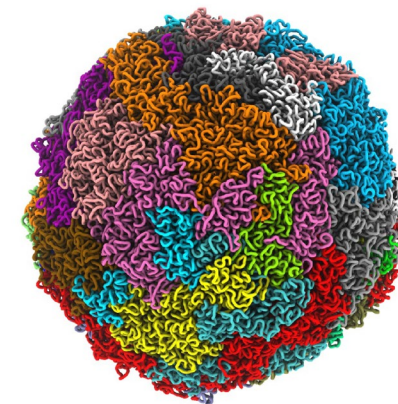
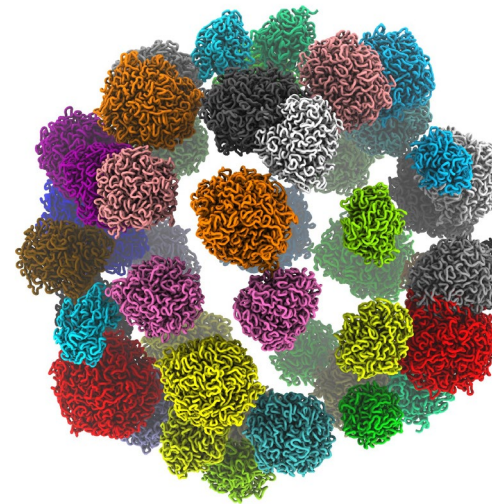
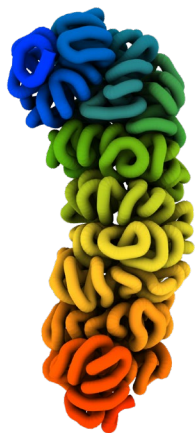
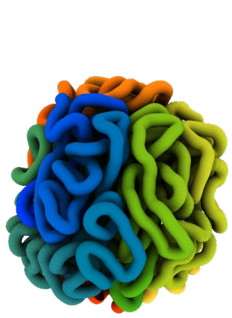
Oliveira A. & Contessoto V et al. JMB, 2020

open-michrom.readthedocs.io

Interphase

prometaphase

metaphase



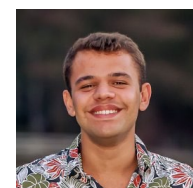
Shaping the genome across the cell cycle



OpenMiChroM



Human full nucleus @50kb resolution



Matheus F. Mello

Modeling *Aedes aegypti* genome

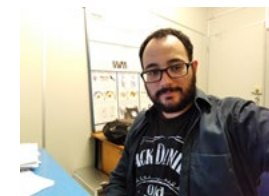


Incorporating motor activity



Exp. Bacteria Hi-C

Pseudomonas aeruginosa

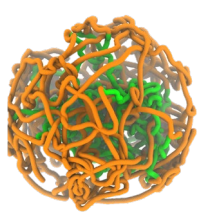


Antonio B. Oliveira Jr

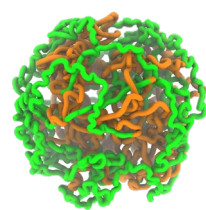


Centromere

Telomeres



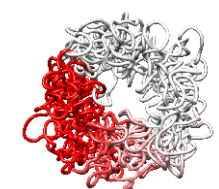
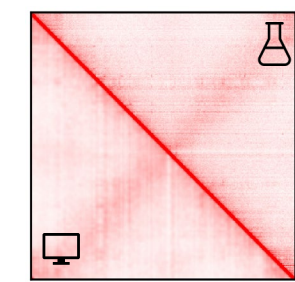
Active



Passive



Sumitabha Brahmachari



Ragini Mahajan

Acknowledgments

José N. Onuchic (Rice Univ)

Peter G. Wolynes (Rice Univ)

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Antonio B. de Oliveira Jr. (Rice Univ)

Esteban Dodero-Rojas (Rice Univ)

Ragini Mahajan (Baylor)

Matheus F. Mello (Rice Univ)

